# docs-pitch-one-pager

marrow — one-pager

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**The embeddable agent runtime for AI that runs where you can't ship Python.** Reference implementation of **ARI**, the open Agent Runtime Interface.

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**Problem.** AI agents are moving into production (57% of teams run agents in prod — *LangChain, 2025*) and escaping the data center — into robots, edge devices, and native apps. Every mature framework (LangChain, LangGraph, CrewAI, AutoGen) is **Python-locked**, and latency is already the **#2 production blocker** (*LangChain, 2025*). You cannot ship a CPython runtime into a real-time robot or a customer's native binary.

**Solution. marrow** is a native **C++17** agent runtime — state, providers, tools, routing, lifecycle — embeddable with **no managed runtime required**, ergonomic from Python. It is the **reference implementation of ARI**, an open standard for the agent *runtime* layer that sits **beneath MCP and A2A** (complementary, not competitive).

**Why it wins (the moat).** ARI's **Embedded profile** is defined by constraints a Python framework *structurally cannot meet* — native core, bounded memory, mandatory cancellation/timeouts, near-zero dependencies. We own both the **standard** and its **reference implementation**, plus a **conformance kit** that makes "ARI-conformant" falsifiable by anyone. Standard + implementation + proof.

**Market (cited, framed honestly).** AI-agents market ~7*Bin*2025*at* 44–50(*syndicated;range*); *humanoidroboticsTAM* * * 38B by 2035** (*Goldman Sachs*); enterprise agentic-AI adoption **33% of apps by 2028** (*Gartner*). Our wedge (embedded/embodied) is the fastest-growing enabling layer — an early-mover window.

**Status (honest).** Pre-alpha (0.1.0). Shipped: ARI spec v0.1, reference implementation, conformance kit, STRIDE threat model + security hardening + CI scanning. **External users: 0.** First deployment target: Tilo (humanoid).

**Traction / Team / Ask.** *Template — replace with real figures, do not present as fact:* design partners `[e.g., 1 robotics + 1 trading]`; founder `[name + background]`; raising `[$1.5M pre-seed, illustrative]` for `[embedded runtime, Tilo integration, a second conformant implementation, standards evangelism]`.

**Links.** Spec: `ARI-SPEC.md` · Strategy: `docs/ari-strategy.md` · Threat model: `THREAT-MODEL.md` · Repo: github.com/bencrooks-dev/marrow